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Bayesian optimization acquisition functions for accelerated search of cluster expansion convex hull of multi-component alloys

Dongsheng Wen¹✉, Victoria Tucker² & Michael S. Titus²

Atomistic simulations are crucial for predicting material properties and understanding phase stability, essential for materials selection and development. However, the high computational cost of density functional theory calculations challenges the design of materials with complex structures and composition. This study introduces new data acquisition strategies using Bayesian-Gaussian optimization that efficiently integrate the geometry of the convex hull to optimize the yield of batch experiments. We developed uncertainty-based acquisition functions to prioritize the computation tasks of configurations of multi-component alloys, enhancing our ability to identify the ground-state line. Our methods were validated across diverse materials systems including Co-Ni alloys, Zr-O compounds, Ni-Al-Cr ternary alloys, and a planar defect system in intermetallic $(\text{Ni}_{1-x}, \text{Co}_x)_3\text{Al}$. Compared to traditional genetic algorithms, our strategies reduce training parameters and user interaction, cutting the number of experiments needed to accurately determine the ground-state line by over 30%. These approaches can be expanded to multi-component systems and integrated with cost functions to further optimize experimental designs.

Materials discovery in today's chemistry, condensed matter physics and materials science relies heavily on the computational simulations that accurately predict the thermodynamically-stable crystal structures at certain finite temperatures. Electronic structure calculations enable the prediction of numerous material properties, such as formation energies and band gaps, with relatively few intrinsic assumptions^{1,2}. However, leading electronic structure calculation theories, such as density functional theory, computationally scale as $O(n^3)$, where n is the number of electrons¹. Despite significant advances in computational power, significant resources are still required to calculate a sufficient number of atomic configurations that enable statistical mechanics calculations^{3–7}, which provide important thermodynamic properties for integrated computational materials engineering (ICME) frameworks. The core of first-principles thermodynamic study is the understanding of how the formation energy is related to the atomic arrangements in the lattice given the composition. Developing the convex hull of the chemical phase field is essential to the synthesis of stable and metastable compounds for novel functional materials^{8,9}.

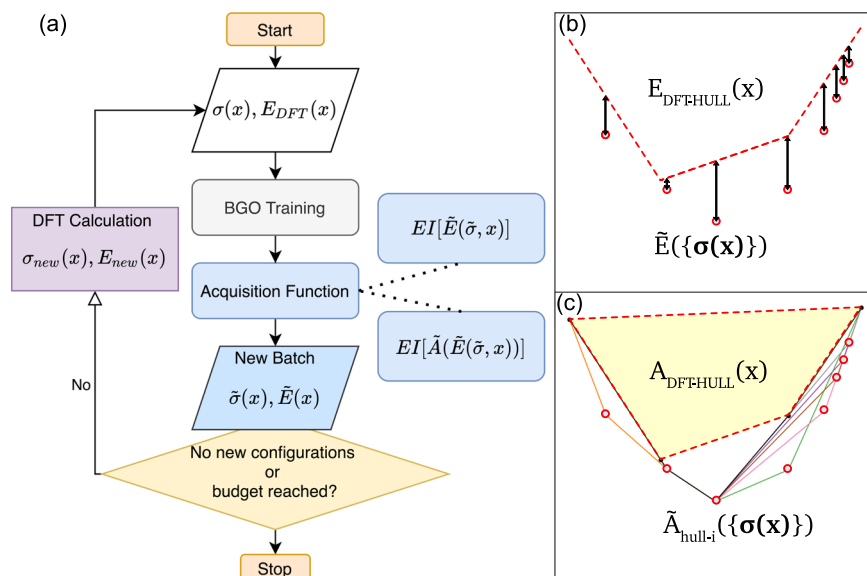
The applications of surrogate models, such as configurational cluster expansion¹⁰, have provided accurate approximations of the energy of the

system with fewer expensive simulation experiments, which have been applied to study phase transformation, order-disorder states, and crystalline defects in multi-component systems^{4–7,11–18}. Cluster expansion methods model the formation energies of a multi-component system by simply assigning lattice site variables within a parent lattice type, which further provides a parameterized effective Hamiltonian that can be directly coupled with statistical ensembles and Monte Carlo simulations to evaluate the thermodynamic properties^{4–7}. For example, first-principles phase diagrams at finite temperatures can be constructed with this routine^{3–5}. It also enables researchers to investigate defect properties and magnetic properties that are difficult to experimentally measure^{6,7,19,20}.

With a data pool of limited structure candidates, the reliability of the cluster expansion depends on how well it is trained by the learning techniques, such as genetic algorithms²¹, direct minimization⁵, compressive sensing²², and Bayesian optimization¹¹. In a typical application, the training data is acquired by selecting the optimal structures to perform first-principles calculations iteratively with an objective function. Taking the advantages of utilizing the uncertainties of predicted quantities, Kristensen et al. has pointed out that searching the ground-state surface (or the convex

¹Department of Chemistry, University of Liverpool, Crown Street, Liverpool, L69 7ZD, UK. ²School of Materials Engineering, Purdue University, 701 West Stadium Ave, West Lafayette, IN, 47907, USA. ✉e-mail: wen94@liverpool.ac.uk

Fig. 1 | Workflow of Bayesian optimization and acquisition functions to search the ground-state line of a configurational space. **a** Bayesian-Gaussian optimization process for learning the ground-state energy. Expected improvement policies for Bayesian-Gaussian optimizations: **(b)** EI of configurations calculated by the distance between the energy to the observed DFT convex hull (dashed line). **(c)** EI of hull area formed by configurations below or on the observed convex hull. Details of the formulations in **(b)** and **(c)** are provided in the Method section “EI of global minimum” – “EI of hull area/volume”.



hull) can maximize the information of the energy-composition space²³. Bayesian optimization with the expected improvement function maximizes the outcome with limited computational budgets²³. With increasingly complicated structures and compositions of a material, it is important to develop acquisition strategies that select the batch experiments with maximum yields.

A Bayesian-Gaussian (BG) optimization process initiates with an initial dataset from DFT calculations to train the BG model, as illustrated in Fig. 1. A Bayesian optimization model was constructed based on the observation that the converged effective cluster interactions (ECI) usually centers around small values^{22,23}, see “Cluster expansion”-“Bayesian optimization” in the Method section. This model employs acquisition function strategies to determine subsequent batches of DFT calculations, aiming to optimize yield. The process concludes upon meeting convergence criteria or exhausting the computational budget, emphasizing the critical role of acquisition functions when resources are constrained.

Acquisition functions in Bayesian optimization prioritize the experiments through the improvement of a target configuration based on the predicted energy and its uncertainty. Such process balances the explorative and exploitative learning to search the ground state line while not losing the accuracy of the model for the whole system²³. The acquisition functions assess the predicted covariance of the configurations in the data pool and provides the optimal batch based on the expected improvement (EI) (“EI of global minimum”-“EI of hull area/volume” in the Method section). A typical covariance matrix from the Bayesian model can be seen in Fig. 2a, which is based on the correlation functions of the configurations. In this study, we propose two new acquisition strategies named as EI-below-hull and EI-hull-area to facilitate the learning process, as shown in Fig. 1b, c. While the computational resource limits the number of observations, EI scores and ranks the batches of subsets of convex hull formed by the combinations of predicted configurations (Fig. 2b), see “EI below hull”-“EI of hull area/volume”. EI-global-min was proposed by Kristensen et al. to evaluate the improvement based on the distance to the global minimum energy²³. Here, the new EI-below-hull measures the improvements by considering the distances of configurations to the convex hull. The EI-hull-area measures the improvements by taking the hull volumes of predicted and observed convex hulls as metrics, as outlined in “EI of global minimum”-“EI of hull area/volume”. This acquisition rank the batches of the configurations based on their contributions to maximize the hull area in a binary system, or volume of the convex hull in a higher dimensional system.

To evaluate the effectiveness of these acquisition functions, we focus on the application of acquisition functions to the convex hull problems of the

Co-Ni alloys⁷, Zr-O oxides⁵, ternary Ni-Al-Cr alloys²⁴, and superlattice intrinsic stacking fault of the $(\text{Ni}_{1-x}\text{Co}_x)_3\text{Al}$ systems, which are usually achieved by genetic algorithms on cluster expansion (named as GA-CE-hull). We demonstrate that the proposed acquisition that maximizes the area/volume of the convex hull can most efficiently explore the convex hull among current acquisition policies without sacrificing the accuracy of the cluster expansion models.

Results

Co-Ni binary alloys

The dataset of the Co-Ni binary system in the stacking fault structure contains 58,165 configurations in the intrinsic stacking fault structure⁷. The cluster expansion energies were trained by 399 DFT-calculated configurations of the system (Fig. 3), where the root-mean-square error and cross-validation score are lower than the DFT convergence criteria of 1 meV/atom⁷. In order to evaluate the performance of the proposed acquisition functions, we use the ground-state line error to measure the difference between the observed convex hull and the target convex hull. The ground-state line error is defined by:

$$\text{GSLE} = \sqrt{\frac{\int_0^1 (\tilde{E}_C(x) - E_T(x))^2 dx}{\int_0^1 E_T^2(x) dx}} \quad (1)$$

in which $E_T(x)$ is the target convex hull, and $E_C(x)$ is the convex hull based on the currently calculated configurations. A lower value of the GSLE indicates a better improvement from the observed data. The target convex hull of the system is shown as the blue line in Fig. 3a. For the four acquisitions policies, the GA-CE-hull, the EI-global-min method, the EI-below-hull method, and the EI-hull-area method, all the experiments start with the same 32 initial data points, see the black square symbols in Fig. 3a, c, e, g. The acquisition functions yield a batch of 5 configurations in maximum for each iteration. It should be noted that the GA-CE-hull and EI-hull-area methods may observe fewer than five configurations in an iteration. For both methods, the predicted on-hull configurations can be fewer than the budget. In addition, the EI-hull-area compares the EI scores of batches and may pick a batch with fewer configurations than the maximum-allowable budget.

The iteration snapshots in Fig. 3 compare the improvements of the ground-state line at iteration-5 and iteration-10 for the methods considered. It can be seen that all the BG-based acquisitions reached the minimum energy across the composition space (−18.25 meV/atom) within five

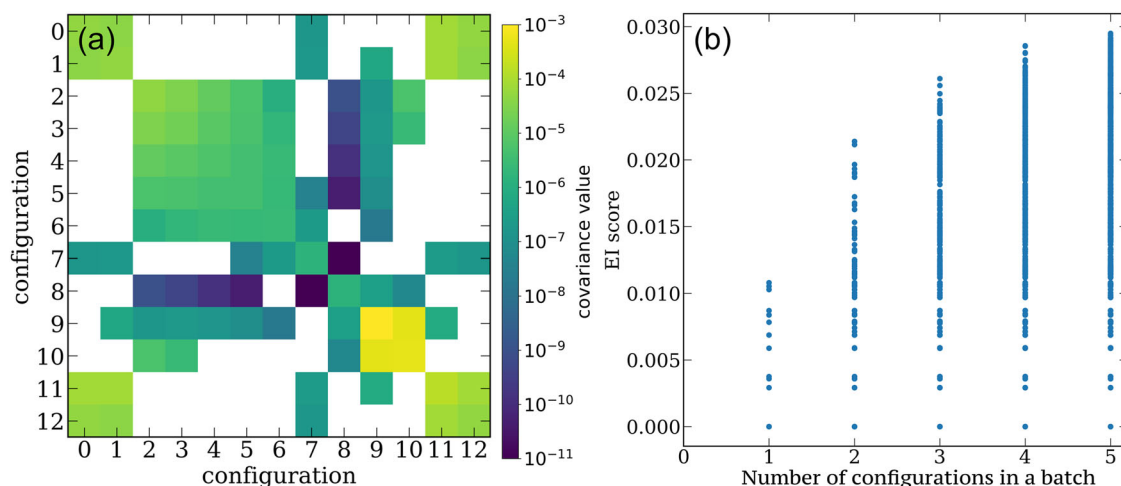


Fig. 2 | Predicted uncertainties of on-hull configurations and the EI scores of the EI-hull-area method. **a** Covariance of 13 on-hull configurations predicted after the first iteration. **b** EI scores calculated by the EI-hull-area method. The scores are

evaluated for various subsets of the predicted hull configurations, each point represents the EI score of a selected subset with the x-axis represents the number of configurations included in the subset.

Fig. 3 | Snapshots of convex hull evolutions and added experiments for different acquisition strategies. GA-CE-hull: (a) iteration-5 and (b) iteration-10. EI-global-min: (c) iteration-5 and (d) iteration-10. EI-below-hull: (e) iteration-5 and (f) iteration-10. EI-hull-area: (g) iteration-5 and (h) iteration-9. Red curves represent the observed convex hulls after each iteration. Blue curves represent the true convex hull. Black squares are initial data points. Red circles are the aquired data points during the learning processes.

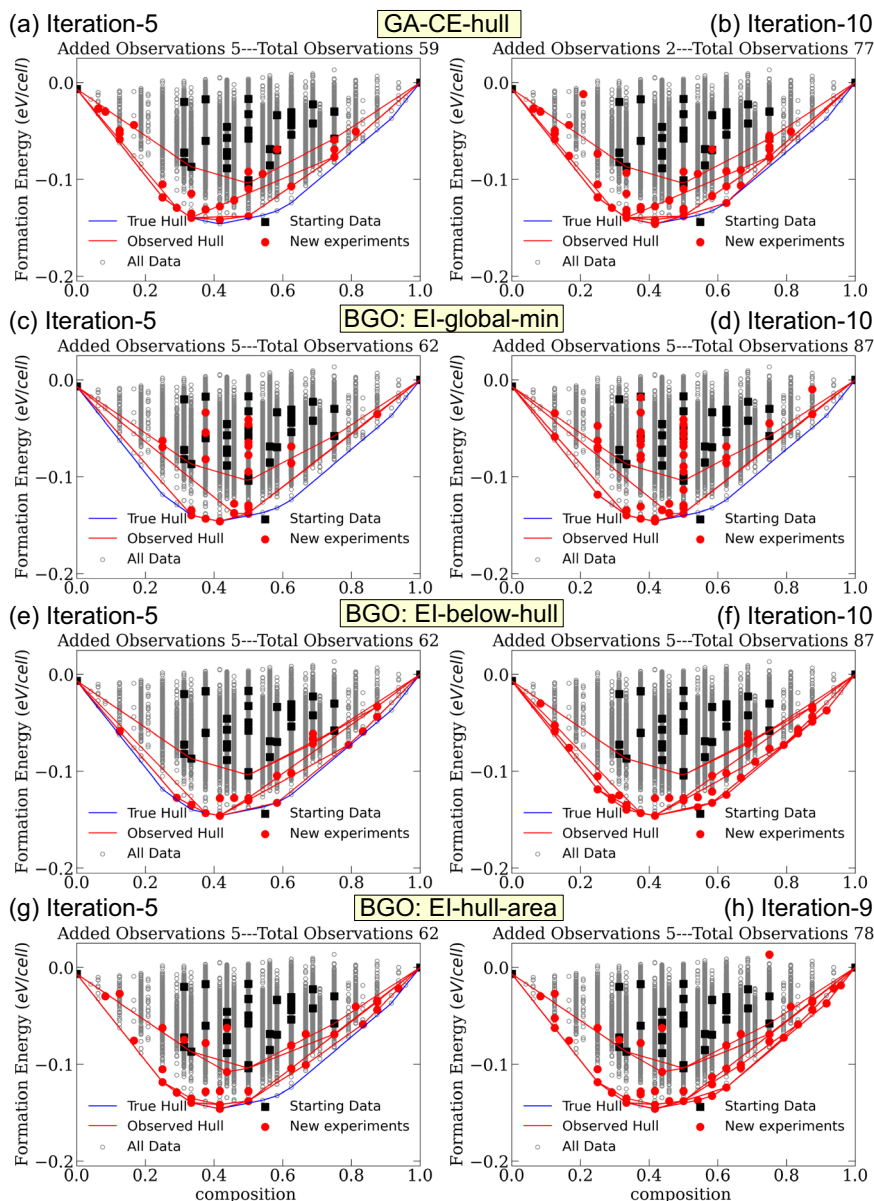


Fig. 4 | Comparison of the acquisition policies when learning the target convex hull of the Co-Ni system. **a** The graph reports the drop of the GSLE with respect to the number of iterations in the learning processes. **b** The total number of observations after each iteration. The error bars are evaluated from 10 independent runs for each method with the same initial training data.

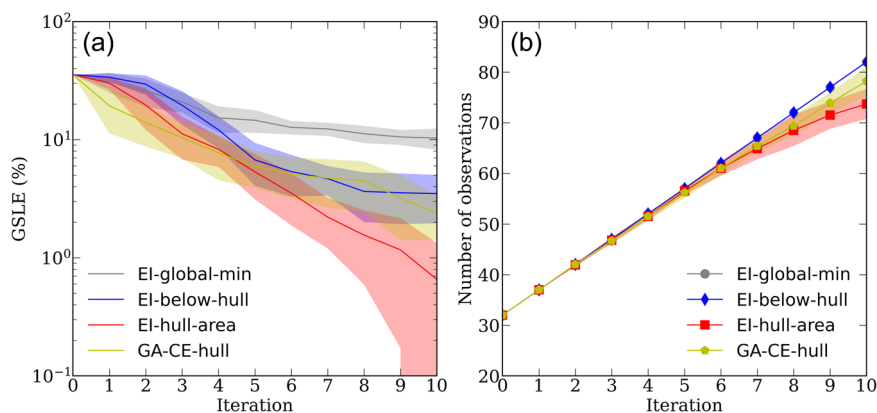
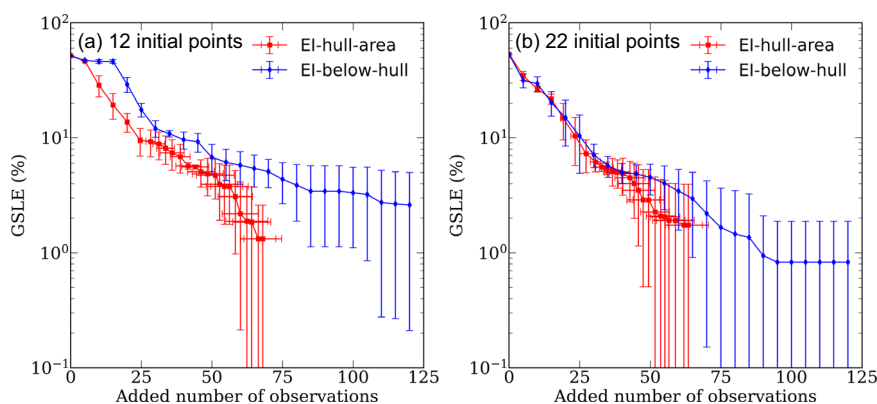


Fig. 5 | The influence of the initial data pools on the learning processes of the EI-below-hull and EI-hull-area acquisition methods. The y-axis is the ground-state line error, the x-axis is the number of added configurations to the initial data. Results are shown for (a) 12 initial data points, and (b) 22 initial data points.



iterations with 62 observations, see Fig. 3c, e, g. Within five iterations, the EI-global-min missed the on-hull structure in the low-concentration range (see Fig. 3d). After iteration-10, GA-CE-hull missed two configurations on the true convex hull with 77 observations. EI-below-hull predicts the target hull while EI-global-min fails (see Fig. 3f, d). The EI-hull-area predicts the target hull after iteration-9 with 78 configurations, see Fig. 3h.

Figure 4a shows the GSLE with respect to the number of iterations. For each method, ten acquisition experiments were carried out to quantify the uncertainty of the learning paths. The improvements of the ground-state line are comparable before the 5th iteration, after which the curves diverge. EI-hull-area shows the greatest improvement, followed by GA-CE-hull and EI-below-hull with comparable errors. The improvement of EI-global-min slows down after six iterations. Figure 4b presents the total number of observations made after each iteration. EI-below-hull and EI-global-min picked five configurations according to their EI scores after each iteration, making 87 observations after ten iterations. On the other hand, as mentioned above, GA-CE-hull and EI-below-hull may make fewer than five observations allowed in a batch. In addition, EI-hull-area made fewer observations than GA-CE-hull after the 7th iteration. Therefore, EI-hull-area shows better learning performance than other methods with fewer observations and lower GSLE.

We noticed that the EI-global-min is less likely to converge within ten iterations. This is expected with the EI formulation in Eq. (21). When the global minimum energy is found, the EI scores of the configuration away from the composition with the minimum energy can be negligible²⁵. This is because the uncertainty of the EI originates from the basis functions of a configuration which cannot be directly related to its composition. On the other hand, for the EI-below-hull method, the EI scores are represented by the distance to the observed composition-dependent convex hull, which allows prioritizing configurations far from the compositions of the global

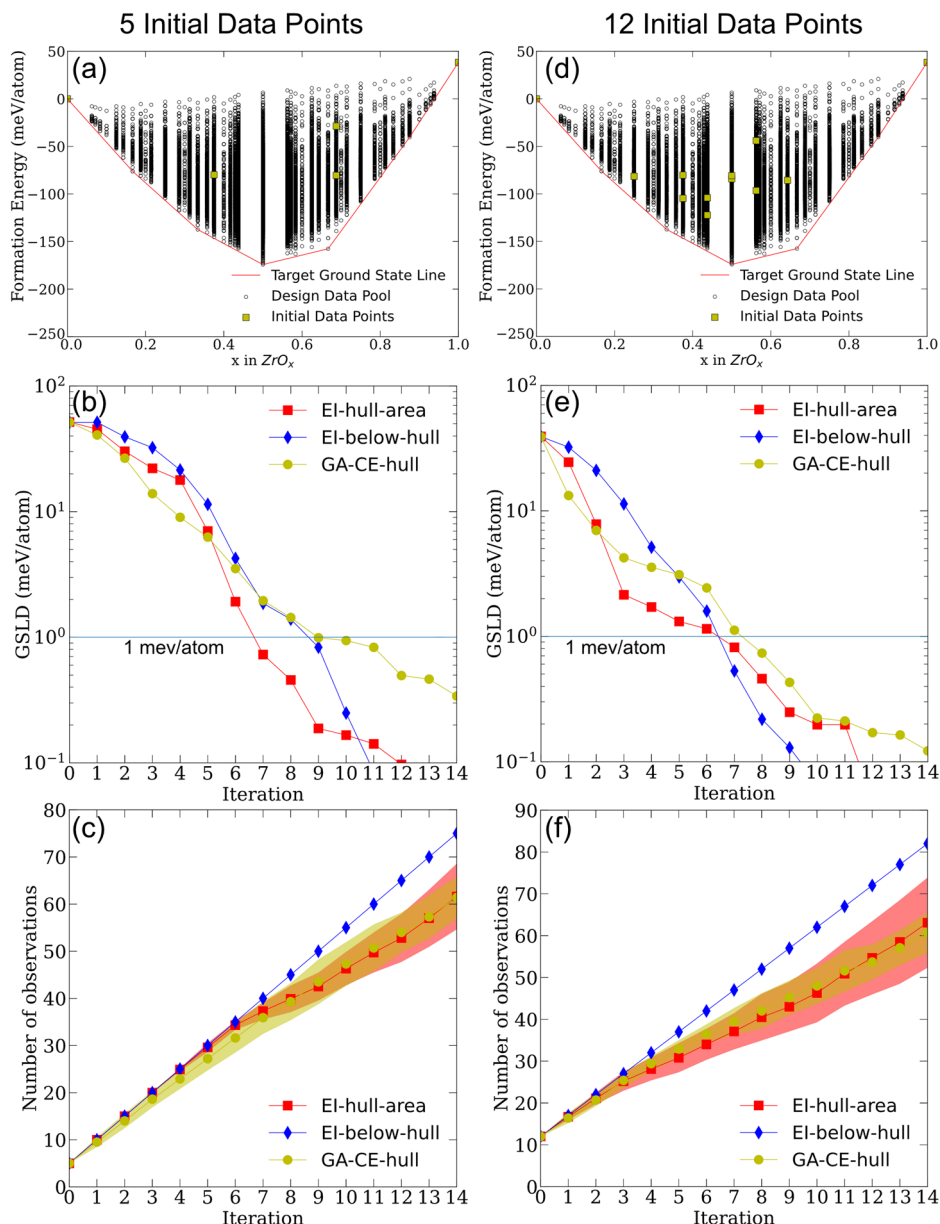
minimum. The EI-hull-area method prioritizes the batch of configuration that maximize the score based on the hull volume defined in the shoelace formula, which also allows sampling configurations across a larger composition range.

Different initial training data will lead to different initial surrogate fits; therefore, the influence of initial data pools on the learning process is studied for the proposed EI-below-hull and EI-hull-area methods. Figure 5 shows the learning results for using 12 and 22 data points to train the initial models. Experiments were carried out with the same acquisition budget that the maximum number of observations made at each iteration is five. The rate of learning the target ground-state line depends on the initial pool, especially for the EI-below-hull strategy. The initial models trained by different data points will explore different regions of the energy-composition space and lead to uncertainties of the learning trajectories. For the EI-hull-area, the learning rates are comparable with 12 and 22 initial data points, and the GSLE drops to about 1% after making about 70 total observations. This is important that one may rely on a smaller size of initial data pool to start the learning.

Zr-O system

The dataset of the Zr-O system contains 93,734 configurations in the α -Zr phase⁵ studied by Puchala and Van der Ven⁵. Figure 6a shows the design data pool and the target ground-state line of the system. The target cluster expansion Hamiltonian was trained by 285 DFT-calculated configurations that the RMSE of the fit was about 3.9 meV/atom⁵. To construct the basis functions, 75 cluster interactions was included by using the interaction cutoffs of 9 Å for pair interactions, 7 Å for triplets, and 6 Å for quadruplets, which are consistent with the published study. For measuring the difference between the target ground-state line and the observed ground-state line, we define the ground-state line difference (GSLD) using the mean

Fig. 6 | Learning the ground-state line of the α -phase ZrO_x system using the EI-hull-area, EI-below-hull, and GA-CE-hull schemes. For five starting training data points: (a) shows the starting data points, design pool, and the target convex hull; (b) shows the evolution of GSLD with training iterations; (c) shows the number of observations with training iterations (the lines and the shaded regions represent the means and the standard deviations from ten acquisition experiments). For twelve starting training data points: (d) shows the starting data points, design pool, and the target convex hull; (e) shows the evolution of GSLD with training iterations; (f) shows the number of observations with training iterations (the lines and the shaded regions represent the means and the standard deviations from ten acquisition experiments).



absolute error:

$$GSLD = \frac{\sum_{x=0}^{N=100} |\tilde{E}_C(x) - E_T(x)|}{N} \quad (2)$$

in which $E_C(x)$ is the currently observed ground-state energy at concentration (x) , $E_T(x)$ is the target ground-state energy at concentration (x) , and $N = 100$ defines 100 evenly-spaced points between $x = 0$ and $x = 1$ for the oxygen concentration in ZrO_x . A well-converged convex hull is expected to exhibit $GSLD \leq 1$ meV/atom, which is about the precision of the DFT calculation⁵.

Two different starting training sets were used to compare the EI-hull-area, EI-below-hull, and GA-CE-hull acquisition schemes described above, using the same training parameters and a batch budget of five configurations for each iteration. The first set starts with five initial data points to train the models, as shown in Fig. 6a. Among the three considered acquisition methods, the GSLD of EI-hull-area converges within 0.8 meV/atom after seven iterations, followed by the EI-below-hull and the GA-CE-hull, which converges using nine iterations, see Fig. 6b. In addition, Fig. 6c shows that the EI-hull-area used fewer observations to reach the target convex hull.

Similar conclusions can be obtained from Fig. 6d, e: using twelve initial starting points, the EI-hull-area scheme converges within eight iterations with less than 40 observations.

Ni-Al-Cr Ternary system

The acquisition algorithms were further extended to study the Ni-Al-Cr ternary system, whose dataset contains 2401 DFT-calculated configurations within the FCC lattice²⁴, see Fig. 7a where the target convex hull is shown by red lines. Three end members and 17 random configurations were selected as the starting training set to compare the learning results of the acquisition functions, see the blue squares in Fig. 7a. For this ternary system, the iterative learning progresses can be captured by comparing the GSLE of the observed hulls and the target hull calculated by Eq. (1) in the barycentric ternary space. Using the same workflows as the aforementioned examples, both the EI-hull-area and EI-below-hull learn the target hull faster than the GA-CE-hull scheme, see Fig. 7b. The EI-below-hull exhibits the steepest learning rate by reaching $GSLE < 2\%$ within 12 iterations, followed by the EI-hull-area. The difference between EI-below-hull and EI-hull-area becomes negligible after 50 iterations (about 300 configurations), where the GSLEs are less than

Fig. 7 | Learning the convex hull of the Ni-Al-Cr system using the EI-hull-area, EI-below-hull, and GA-CE-hull schemes. **a** DFT formation energies (grey points) and the ultimate convex hull (red lines) of 2401 configurations. The blue squares represent the initial training data points and the blue lines form the initial hull. **b** shows the ratio between the observed hull volume and the true hull volume with increasing number of observations using three learning schemes.

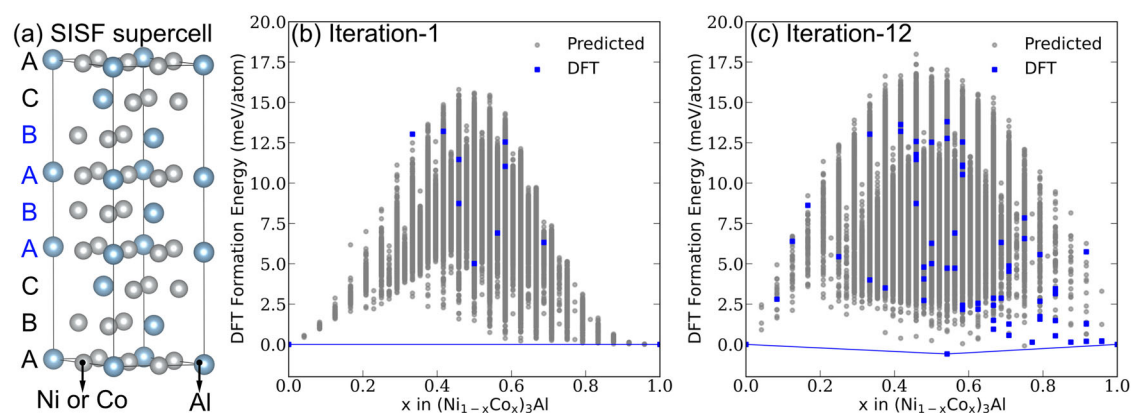
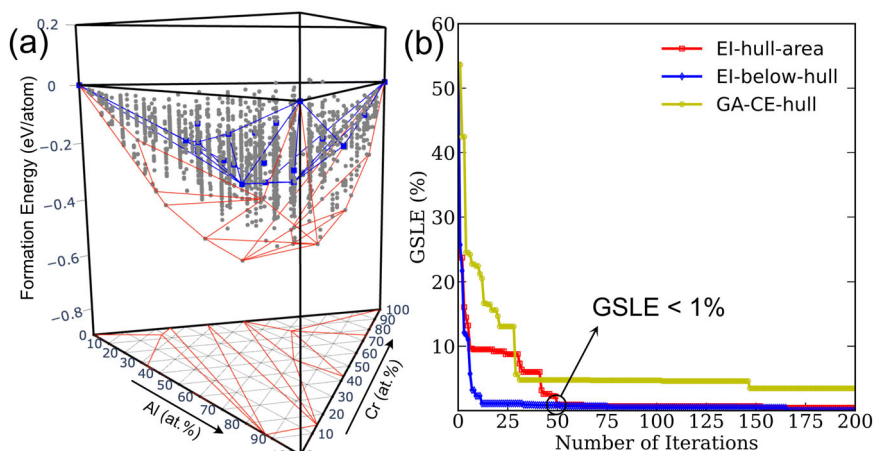


Fig. 8 | Exploring the convex hull of $(\text{Ni}_{1-x}\text{Co}_x)_3\text{Al}$ in superlattice intrinsic stacking fault structure. **a** Supercell prototype of the SISF $(\text{Ni}_{1-x}\text{Co}_x)_3\text{Al}$, containing a stacking sequence of (...ABCABABCA...) with a SISF in the middle. Both Co or Ni are allowed to occupy the silver atomic sites and Al occupies the blue atomic

sites. **b** During iteration-1, the DFT-calculated energies (blue) are above the hull, resulting in a BG model with no predicted energies below the DFT hull. **c** After iteration-12, the EI-global-min method observed the configuration breaking the convex hull.

0.5%. The learning process of the GA-CE-hull shows limited improvements after 30 iterations.

Exploring the convex hull of $(\text{Co}, \text{Ni})_3\text{Al}$ superlattice intrinsic stacking fault

The proposed acquisition functions can be useful to explore the convex hulls of structures with low symmetries, where millions of configurations easily exceed the computational power⁷. For example, superlattice intrinsic stacking faults (SISFs) in the $L1_2$ phase of $(\text{Co}_x, \text{Ni}_{1-x})_3\text{Al}$ can be formed during high-temperature creep deformation stages²⁶, breaking the symmetry by changing the (111) plane stacking sequence from (...ABC...) to (...ABCABABCA...), see Fig. 8a. Understanding the elemental segregations to SISFs in Ni- and Co-based superalloys is important to design novel superalloys with improved high-temperature creep performance^{6,7,27–30}. Cluster expansion surrogate models based on DFT calculations can facilitate the alloy design through calculating the temperature- and composition-dependent segregation profiles and stacking fault energies^{6,7}. However, within this 32-atom supercell in Fig. 8a, allowing the configurational degree of freedom of Co on Ni sites leads to millions of symmetrically-distinct configurations, which require extremely large memory and storage to handle the data. Therefore, the proposed acquisition schemes are applied to explore the convex hull of the $(\text{Co}_x, \text{Ni}_{1-x})_3\text{Al}$ SISF system. To mitigate the large configurational space, a truncated set of total number of 99,361 configurations were enumerated randomly for the SISF structure in $(\text{Co}_x, \text{Ni}_{1-x})_3\text{Al}$. During the first iteration, this EI-hull-area could not be used directly used due to the fact that the initial model based on positive DFT energies did not predict configurations

breaking the flat convex hull, see Fig. 8b. However, this can be overcome by the EI-global-min and EI-below-hull methods, which are competent to search global minimum energies and does not rely on a well-defined convex hull of the predicted energies. After twelve iterations, the EI-global-min successfully found the current convex hull with one configuration breaking the flat ground-state line, which is shown in Fig. 8c. This application also demonstrates the advantages of using different acquisition functions to explore the convex hull of low-symmetry structures.

Validation

Fewer observations do not necessarily mean a loss of information in the cluster expansion²³. We further validate the cluster expansions by calculating the crucial thermodynamic properties of the FCC Co-Ni alloys reported in the previous study⁷. The finite-temperature free energies and heat capacities can be evaluated by semi-grand canonical Monte Carlo simulations (GCMC) using the cluster expansion effective Hamiltonian^{5,7}. In the previous study, the cluster expansion model was trained with a training set of 629 configurations calculated by DFT⁷. The Gibbs free energy (G), entropy (S), and enthalpy (H) of the $\text{Ni}_{40}\text{Co}_{60}$ alloy were calculated by free energy integrations using the CASMcode software³¹, see Fig. 9(a). Following the same procedure as the previous work⁷, GCMC simulations and thermodynamic calculations were carried out based on three cluster expansion models trained by different number of observations: (1) Training set A contains 80 configurations including the convex hull; (2) training set B contains 315 configurations including the convex hull; and (3) training set C contains all the 629 configurations. The heat capacities of $\text{Ni}_{40}\text{Co}_{60}$

Fig. 9 | Thermodynamic properties of FCC Ni₄₀Co₆₀ alloy. **a** Gibbs free energy (G), entropy (S), and enthalpy (H) of the Ni₄₀Co₆₀ alloy reproduced from the previous study⁷. **b** Calculated heat capacity C_p of the Ni₄₀Co₆₀ alloy based on the cluster expansion models trained by three sets of training data.

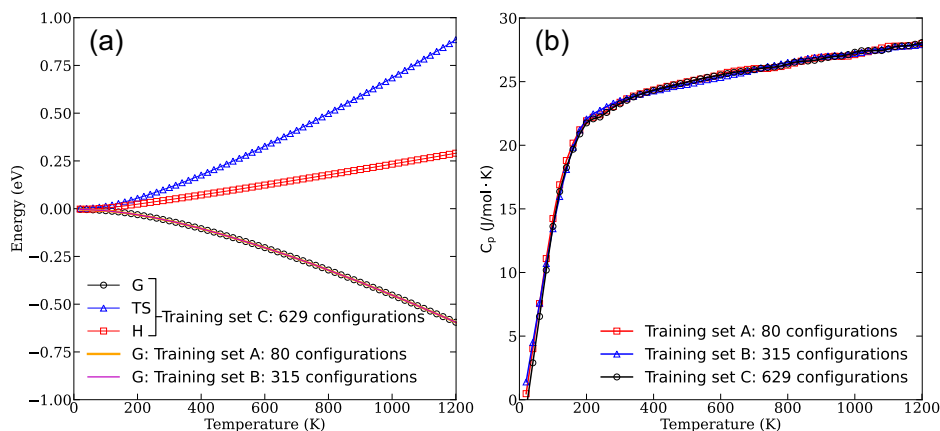


Fig. 10 | Confusion matrices of the BG models for predicting the on-hull configurations in the Co-Ni, Zr-O, and Ni-Al-Cr systems at the end of the training processes using the EI-hull-area and EI-below-hull acquisition functions.



evaluated by these three models are in excellent agreement (see Fig. 9(b)), indicating that significant effective interactions can be captured by a truncated set of configurations. This observation further justifies the application of BG models on fast exploration of the convex hull of an alloy system. Because a typical DFT calculation may take hours to days to finish, an optimal strategy can facilitate the materials discovery processes by reducing hundreds of hours of DFT simulations.

Discussion

Both EI-hull-area and EI-below-hull methods take the advantages of Bayesian optimization to rank a batch of simulation experiments based on both the improvement and the uncertainty of the predicted hull configurations. These new acquisition functions outperform the traditional scheme with genetic algorithms in searching the ground-state line for alloying and ionic systems. This can be attributed to the fact that multiple experiments may fulfill exploration and exploitation of the Bayesian model at the same time. In addition, Bayesian optimization does not rely on selecting the best set of ECIs when assuming the Gaussian distribution for the ECIs²³, significantly simplifying the fitting procedures and reducing user interactions. On the other hand, genetic algorithms can trap in the local minima; additional direct minimizations are often necessary to find an optimal ECI set, which requires more user knowledge to weigh the training data with fine-tuning parameters^{5,7}. The EI-global-min method developed

in the previous study is effective in searching the global minimum iteratively from one composition to another²³. However, EI-global-min is less competitive in selecting batch experiments. This is because EI-global-min is likely to sample the configurations with similar compositions that yield high EI scores, as shown in Fig. 6. EI-below-hull and EI-hull-area are more explorative on the compositions, which further empower the data acquisition processes through batch selections to optimize the parallel calculation throughputs on supercomputers. The BG models at the end of the acquisition processes of EI-below-hull and EI-hull-area show excellent convergence on the observed data and on-hull configurations, as shown in Fig. 10, due to the fact that BG processes maximize the Gaussian likelihood of making the observations on the energies³².

The new acquisition functions are readily extended to ternary and multi-component systems. In ternary systems, the EI in the EI-below-hull will be calculated by replacing the 2d convex hull $E_N(x)$ with the 3d energy convex surfaces. For the EI-hull-area, the area quantity will be replaced with the volume of the convex hull, which can be calculated with the triangulation and the tetrahedron volumes³³. For N-component systems (N larger than 3), the composition formation energy convex hull is an irregular (N-1)-dimensional polytope where the vertices of on-hull configurations defines its volume (see “EI of hull area/volume” in the Method). While the EI-below-hull scheme relies on the distance between a configuration and the facet of the polytope, EI-hull-area seeks to expand the volume of the polytope. In larger systems,

these methods enable direct searches without starting from binary subsets like traditional routines. In addition, the EI-below-hull and EI-hull-area can be properly incorporated with cost functions to further optimize the computation efficiency. For example, the computation cost of each batch can be roughly evaluated by the number of configurations in each batch, such as Fig. 2b. In addition, the cost scales with the number of atoms/electrons of each configuration, and therefore, the computation time for a batch may be related to the largest or the most complex structure. A proper cost function will be meaningful by considering the computation time in the parallelization, which depends on both the available nodes and CPUs per node in a practical scenario. Kristensen et al. have employed the cost function to evaluate the cost of every configuration that the Pareto front can be developed for making decisions²³. Future work will focus on such schemes to maximize the output of the acquisitions.

Applications of acquisition functions depend on the complexity of the energy hull, and one may use different techniques to explore. For a convex hull in the composition-energy space, EI-hull-area is formulated to maximize the convex hull volume; therefore, it tends to select both the highest positive and the lowest negative energies to expand the hull in both directions. While high-energy configurations are valuable to train and build high-temperature free energy models, the bottom-hull is more important to learn the ground-state configurations. Therefore, in the current study, the positive formation energies are excluded from the hull. In addition, when the predicted hull reaches the true convex hull, it is likely that the improvement by the predicted hull volume based on Eq. (23) can be negligible. In this case, the improvement function can be replaced by $\max[0, A_N]$, where A_N is formed by the configurations slightly above the observed hull within a tolerance threshold. For a concave hull scenario, the systems possess a positive formation energy landscape (configurations are likely above the tie line/facet between end members). In this case, the improvement in EI-hull-area may be reconstructed with the $\min()$ function to replace Eq. (23) to minimize the volume between the concave hull and the end-member facet. However, such systems are less interesting because they are likely thermodynamically unstable and decompose to end members at elevated temperatures.

A random sampling of configurations to initiate the search does not show significant influence on the learning results of the tested systems, indicating that one can rely on a smaller set of initial batches to reduce the computational cost. Interestingly, Ober and Van der Ven showed that the energy extrema of the cluster expansion may be located at the vertices of the high-dimension ECI space³⁴. While the size of the ECI space depends on the truncated interactions of clusters, highly ordered structures are located at the vertices of the ECI space within nearest-neighbor interactions³⁴. Therefore, initial sampling may maximize the information to train the models when selecting those configurations at or near the vertices of the polyhedron formed by the nearest-neighbor correlation functions, which will be studied in the future.

We investigated the learning performance of the new acquisition functions based on the expected improvement policy in Bayesian Gaussian optimization models. By learning the convex hull of the Co-Ni, Zr-O, Ni-Al-Cr, and $(\text{Ni}_{1-x}\text{Co}_x)_3\text{Al}$ systems, we demonstrated that the EI-below-hull and EI-hull-area methods outperform the EI-global-min scheme and the traditional GA-CE-hull scheme in searching the convex hull with batch

experiments. With the correct convex hull, we found that the cluster expansion trained by fewer observed data points can reproduce the thermodynamic properties of the Co-Ni system. The proposed acquisition functions are useful to study low-symmetry structures with a large configurational degree of freedom that we plan to use the proposed methods to investigate the phase stability of multi-component alloy systems with complex structures. For future development, we plan to incorporate cost functions to evaluate the costs of batch calculations, which will be important for making decisions.

Methods

Cluster expansion

The energy of a symmetrically distinct atomic arrangement in the multi-component crystalline system can be approximated by the cluster expansion (CE)^{4,10}:

$$E(\sigma(x)) = \sum_i V_i(\sigma(x))J_i \quad (3)$$

in which σ stands for the vectorized components of atomic interactions within the geometry of the crystal, $V_i(\sigma(x))$ represents the basis functions of a collection of symmetrically distinct configurations, J_i represents the non-zero coefficients of the effective cluster interaction (ECI), and the alloy composition is x for a given configuration. The basis function V_i for a configuration is defined by:

$$V_i(\sigma(x)) = \frac{1}{m} \sum_{\gamma \in i} \psi_{\gamma}(\sigma(x)) \quad (4)$$

where i represents a collection of symmetrically equivalent clusters under a space group operation of the lattice sites, γ corresponds to the equivalent clusters defined by i , m is the multiplicity of the equivalent clusters, and $\psi_{\gamma}(\sigma(x))$ for each equivalent cluster is described by the polynomials:

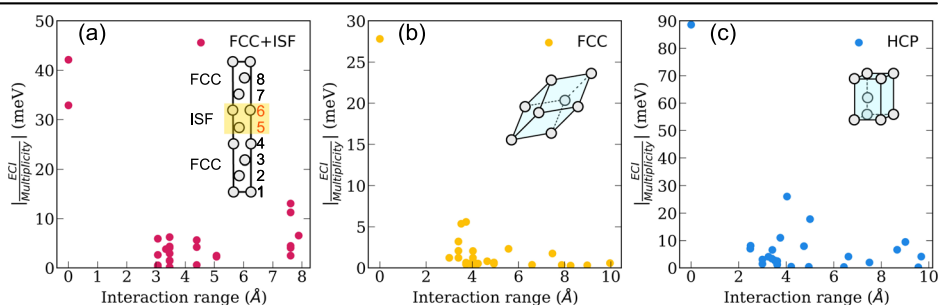
$$\psi_{\gamma}(\sigma(x)) = \prod_{j \in \gamma} \sigma_j \quad (5)$$

where σ_j is the element occupation variable assigned to a lattice site j inside the interacting cluster. The dimension of the basis functions is the number of cluster interactions defined by i , which depends on (1) the type of interactions, such as points, pairs, and triplets, and (2) the maximum length of the interacting sites. The interactions can have multiple equivalent clusters due to the symmetry of a crystal, and the reduced ECI can be expressed by J_i/m where m is the multiplicity of the interaction⁴. For a set of configurations $\Phi = \{V(\sigma)\}$, the energies can be expressed in the matrix form:

$$E(\{\sigma(x)\}) = \Phi J \quad (6)$$

Such representation of the energy turns the complicated quantum mechanical calculations into linear regression problems, significantly reducing the computational costs. Although there exist infinite cluster interactions in a crystal, the energy of the system converges rapidly with a

Fig. 11 | The absolute value of the ECIs decreases with increasing interaction range of clusters different structures of the binary Co-Ni system. a The intrinsic stacking fault of the face-centered cubic structure (FCC+ISF), the inserted graph shows the structure of the supercell. **b** FCC structure. **c** HCP structure.



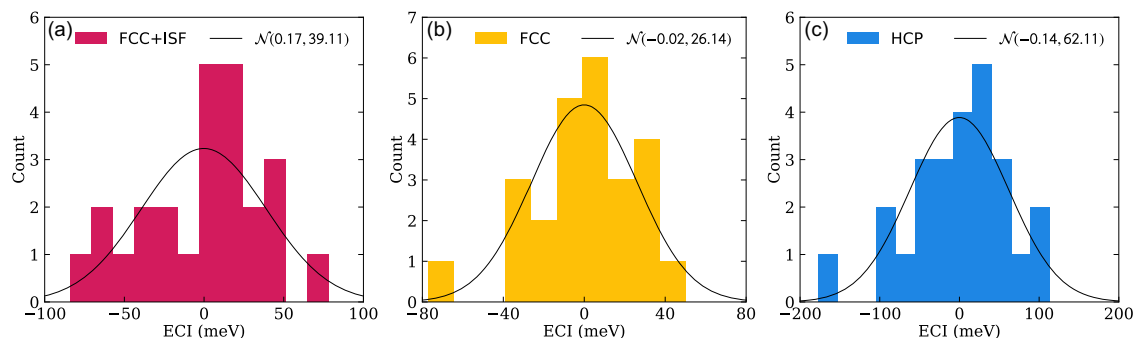


Fig. 12 | Effective cluster interactions of well-converged genetic algorithms. Histograms and Gaussian fittings of the well-converged ECIs for the (a) FCC+ISF structure, (b) FCC structure, and (c) HCP structure of the Co-Ni binary system⁷.

truncated size of ECIs^{4,5,7}. Figure 11 shows the well-converged ECIs of a previous study to investigate the energy of the Co-Ni binary alloys, which can represent the DFT-calculated energy with an uncertainty below one meV/atom⁷. It can be seen that the ECI values generally decrease with increasing distances between the interacting atoms. Therefore, an accurate surrogate model can be obtained by training the ECIs using the ab initio data and learning algorithms⁴.

Bayesian optimization

The key problem is to learn the ECI with as few expensive simulations as possible. Compared to other machine learning techniques, Bayesian optimizations coupled with the Gaussian process have the advantages of quantifying the uncertainties of the predicted CE energies and propagate the uncertainties to thermodynamic modelling^{11,22,23,32,34}. The Bayes' theorem can be used to describe the posterior of the ECI by ref. 23:

$$p(J|D_N, \alpha, \beta) \propto p(J|\alpha)p(D_N|J, \beta) \quad (7)$$

in which the prior, $p(J|\alpha)$, defines the probability distribution of the ECI, and $p(D_N|J, \beta)$ is the likelihood of measuring the energies of a collection of N configurations D_N . Instead of carefully selecting a compact set of configuration-important ECIs that predicts the energies with an objective function, such as mean square error and cross-validation^{35,36}, the Gaussian description of the optimal set of ECIs can have as many interactions as possible while the values of the ECIs are constrained to be small^{11,22,23}:

$$p(J|\alpha) = \mathcal{N}(J|0, \alpha) \quad (8)$$

Figure 12 presents the histograms and Gaussian distribution of the value of ECIs for the Co-Ni alloys in the FCC+ISF, FCC, and HCP structures. The Gaussian treatment may be justified by the well-converged ECIs. For the well-converged truncated sets of ECIs (as shown in Fig. 11) trained by the genetic algorithms and the direct minimization methods, the 20-30 selected ECIs were carefully chosen from a group of more than 100 interactions⁷. The histograms of the ECIs may follow the Gaussian distribution centered at zero, see Fig. 12(a)–(c) for the reported mean and standard deviation of the Gaussian fitting. This representation enables the use of the convenient radial basis function kernel method to describe the distributions of ECIs and evaluate the variance³². Therefore, given independent measurements of each configuration in D_N , $p(J|D_N, \alpha, \beta)$ follows the Gaussian distribution^{23,32}:

$$p(J|D_N, \alpha, \beta) = \mathcal{N}(J|m_N, S_N) \quad (9)$$

where m_N represent the values of ECI, and S_N represent the corresponding covariance matrix, which can be expressed as:

$$m_N = \beta S_N \Phi^T E \quad (10)$$

$$S_N^{-1} = \alpha I + \beta \Phi^T \Phi \quad (11)$$

where I is the identity matrix, E is the set of observed energies, and Φ is the matrix form of the basis functions $\{V(\sigma)\}$.

The posterior depends on the hyperparameters (α, β) that are unknown at the beginning. Therefore, the Bayes' rule can be used evaluate the joint posterior based on the experimental observations:

$$p(\alpha, \beta, J|D_N) = p(J|D_N, \alpha, \beta)p(\alpha, \beta|D_N) \quad (12)$$

$$p(\alpha, \beta|D_N) \propto p(D_N|\alpha, \beta)p(\alpha, \beta) \quad (13)$$

where $p(D_N|\alpha, \beta)$ is the marginal likelihood^{23,32}. For learning the cluster expansion and making observations, α and β are independent from each other because α describe the uncertainty of the ECIs while β describe the uncertainty of making an observation. In fact, β depends on how accurate an observation one can expect from the first-principles calculations; therefore, β can be fixed to the convergence criteria of the DFT calculations, $\tilde{\beta}$. In this case, $p(\alpha, \beta)$ may be assumed as a constant such that Eq. (13) can be further simplified to the problem of maximizing the marginal likelihood³²:

$$p(\alpha, \beta|D_N) \propto p(D_N|\alpha, \beta) = \int p(D_N|J, \beta)p(J|\alpha)dJ \quad (14)$$

In this way, the value of α can be evaluated through maximizing the marginal likelihood $p(D_N|\alpha, \beta)$ using the expectation-maximization algorithm³². It can be seen that the $p(D_N|\alpha, \beta)$ marginalizes J and therefore, the ECIs values can be analytically evaluated²³. But this does not affect the prediction because this is through posterior predictive assessment^{32,37}. Expectation-maximization processes determine the optimized hyperparameter $\alpha = \tilde{\alpha}$ such that posterior Eq. (11) can be evaluated. To make prediction, the predicted energy of a configuration follows the Gaussian³²:

$$p(\tilde{E}|V(\tilde{\sigma}), D_N) = \mathcal{N}(\tilde{E}|V(\tilde{\sigma})m^T, \tilde{\beta}K_*) \quad (15)$$

in which $K_* = \kappa(V(\sigma), V(\tilde{\sigma}))$ is the covariance function that evaluates the covariance between the basis functions of the observed data points $V(\sigma)$ and new data points $V(\tilde{\sigma})$. This can be realized by the equivalent kernel in the form³²:

$$\begin{pmatrix} E \\ \tilde{E} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} E_N \\ \mu_E \end{pmatrix}, \begin{pmatrix} K & K_* \\ K_*^T & K_{**} \end{pmatrix}\right) \quad (16)$$

where K is the covariance matrix of the observed data and K_{**} is $\kappa(V(\tilde{\sigma}), V(\tilde{\sigma}))$ ³². Then the predictive mean and variance of the new

configurations based on the observed energies E_N can be expressed by using the equivalent kernel with the Woodbury matrix (K_y)^{32,38}:

$$\mu_E = K_*^T K^{-1} E_N \quad (17)$$

$$\nu_E = K_{**} - K_*^T K_y^{-1} K_* \quad (18)$$

Here, we will introduce three different acquisition functions based on the expected improvement (EI) policy for the BG optimization to learn the ground-state line of the formation energy. In order to compare different strategies, the traditional acquisition strategy for the genetic algorithm will be introduced.

EI of global minimum

The expected improvement policy has shown its efficiencies in searching the true convex hull through maximizing the improvements with respect to global minima^{23,25}:

$$EI(\tilde{E}, \tilde{\sigma}, x) = \mathbb{E}[\mathbf{I}(\tilde{E}, \tilde{\sigma}, x) | D_N, \tilde{\sigma}, x] \quad (19)$$

$$\mathbf{I}(\tilde{E}, \tilde{\sigma}, x) = \max[0, E_N - \tilde{E}(\tilde{\sigma}(x))] \quad (20)$$

in which a configuration can ‘improve’ the current convex hull when its predicted energy, $\tilde{E}(\tilde{\sigma}(x))$, is lower than the observed minimum, E_N . By integrating the distribution of the predicted energy up to the observed minimum E_N , the EI score is computed by:

$$EI(\tilde{E}, \tilde{\sigma}, x) = (E_N - \mu_E(\tilde{\sigma}(x)))CDF\left(\frac{E_N - \mu_E(\tilde{\sigma}(x))}{\nu_E(\tilde{\sigma}(x))}\right) + \nu_E(\tilde{\sigma}(x))PDF\left(\frac{E_N - \mu_E(\tilde{\sigma}(x))}{\nu_E(\tilde{\sigma}(x))}\right) \quad (21)$$

where $\mu_E(\tilde{\sigma}(x))$ is the predicted mean of the configuration by the BG model, $\nu_E(\tilde{\sigma}(x))$ is the predicted variance, $CDF()$ and $PDF()$ are the cumulative distribution function and the probability density function, respectively. The configuration with the highest EI score will be chosen as the next observation point.

In a real-life application of the global EI policy, because the energy is linked to the composition through the configuration-dependent basis functions, the EI scores are usually measured with respect to the σ -space by fixing the composition- x ²³, limiting its yield in the dense energy-composition space. In addition, one would maximize the use of the available computation resources by choosing and calculating a batch of configurations with different compositions. Therefore, the EI acquisition policy should be properly extended to include the composition of a configuration.

EI below hull

A simple modification of the EI of a configuration substitutes the global minimum (E_N) with a convex hull function $E_N(x)$ in Eq. (20) and Eq. (21), leading to:

$$EI(\tilde{E}, \tilde{\sigma}, x) = (E_N(x) - \mu_E(\tilde{\sigma}(x)))CDF\left(\frac{E_N(x) - \mu_E(\tilde{\sigma}(x))}{\nu_E(\tilde{\sigma}(x))}\right) + \nu_E(\tilde{\sigma}(x))PDF\left(\frac{E_N(x) - \mu_E(\tilde{\sigma}(x))}{\nu_E(\tilde{\sigma}(x))}\right) \quad (22)$$

The convex hull function is obtained by interpolating the currently-observed convex hull configurations, see red dashed line in Fig. 1b. The EI of a configuration will be based on its distance to the DFT convex hull evaluated at its composition, see Fig. 1b. This acquisition can yield a batch of configurations with different compositions according to the ranking of the EI scores.

EI of hull area/volume

The increasing complexity of the system can lead to many on-hull configurations, and it is necessary to search the ground-state convex hull since the thermodynamic property is based on accurate reference energy at low

temperatures^{5,7,23}. For example, accurate construction of the Zr-O phase diagram depends on searching the ground-state configurations reflecting the ordering of oxygen in different Zr structures⁵. Iteratively selecting one configuration to observe cannot optimize the workflow. Selecting a batch of dozens of configurations will be redundant when the computational resources are limited. Therefore, we introduce a new algorithm to evaluate the expected improvement of a batch based on the convex hull of the batched configurations. After a BG model is trained with the up-to-date data points, the energies are predicted for the design data pool. The possible convex hull configurations of the predicted energies can be obtained, see the circles in Fig. 1c. Given a collection of n possible new hull configurations $\{x_i, E_i\}$, the number of combinations of subsets containing k configurations is: $\binom{n}{k} = n!/(n-k)!k!$. The improvement for a given subset of k configurations is calculated by comparing the hull area of the subset and that of the current DFT hull:

$$\mathbf{I}(\tilde{A}, \tilde{\sigma}, x) = \max[0, A_N - \tilde{A}(\tilde{E}(\tilde{\sigma}(x)))] \quad (23)$$

in which A_N is the area formed by the current DFT convex hull (highlighted by yellow area in Fig. 1c), $\tilde{A}(\tilde{E}(\tilde{\sigma}(x)))$ is the area of the selected subset of on-hull configurations (highlighted by different line colors in Fig. 1c). In the energy-composition space, the area of a set of configurations can be calculated based on the compositions and energies $\{x_i, E_i\}$ using the shoelace formula³⁹:

$$A = \frac{(\sum_{i=1}^{n-1} x_i E_{i+1} + x_n E_1) - (\sum_{i=1}^{n-1} x_{i+1} E_i + x_1 E_n)}{2} \quad (24)$$

in which x_i is the composition and E_i is the energy of a configuration. This can be rearranged and simplified to:

$$A = \frac{\sum_{i=1}^{n-1} (x_{i-1} - x_{i+1})E_i + (x_{n-1} - x_1)E_n + (x_n - x_2)E_1}{2} = \frac{\sum_i^n \Delta x_i E_i}{2} \quad (25)$$

The uncertainties of the configurations can be propagated to the area of the selected hull subsets through the multivariate Gaussian process³². For a single configuration, the predictive mean and predictive variance are evaluated with the normal distribution²³:

$$p(\tilde{E}|\{\tilde{\sigma}, x\}) \sim \mathcal{N}(\mu_E(\tilde{\sigma}, x), \nu_E(\{\tilde{\sigma}, x\})) \quad (26)$$

The area of the convex hull will also follow the normal distribution:

$$p(\tilde{A}|\{\tilde{E}, \tilde{\sigma}, x\}) \sim \mathcal{N}(\mu_A(\{\tilde{E}, \tilde{\sigma}, x\}), \nu_A(\{\tilde{E}, \tilde{\sigma}, x\})) \quad (27)$$

$$p(\tilde{A}|\{\tilde{E}, \tilde{\sigma}, x\}) \sim \mathcal{N}\left(\frac{\sum_i^n \Delta x_i \mu_{E,i}}{2}, \sum_{i=1}^{i=n} \sum_{j=1}^{j=n} \frac{\Delta x_i \Delta x_j \text{cov}[\tilde{E}_i, \tilde{E}_j]}{4}\right) \quad (28)$$

in which $\mu_A(\{\tilde{E}, \tilde{\sigma}, x\})$ is the predictive mean area and $\nu_A(\{\tilde{E}, \tilde{\sigma}, x\})$ is the predictive variance of the area. The expected improvement of the area then becomes:

$$EI(\tilde{A}, \{\tilde{\sigma}, x\}) = (A_N - \mu_A(\tilde{\sigma}(x)))CDF\left(\frac{A_N - \mu_A(\tilde{\sigma}(x))}{\nu_A(\tilde{\sigma}(x))}\right) + \nu_A(\tilde{\sigma}(x))PDF\left(\frac{A_N - \mu_A(\tilde{\sigma}(x))}{\nu_A(\tilde{\sigma}(x))}\right) \quad (29)$$

The covariance between configuration- i and configuration- j , $\text{cov}[\tilde{E}_i, \tilde{E}_j]$, is essential to propagate the uncertainties of configurations to the hull area calculation. For example, Fig. 2a shows the covariance matrix of 13 configurations from the predicted hull. Calculating all the predicted on-hull configurations may be unrealistic because the batch size depends on real-life situations, such as the number of parallel experiments on supercomputers. If we set the batch size limit to five configurations, there can be many subsets of

convex hulls based on the combinations of the configurations. For each batch, the area of the convex hull can be calculated by the shoelace formula with the compositions and predictive means. And the predictive variance of the energy can be propagated to the variance of the hull area. Figure 2(b) shows the EI scores of the hull areas formed by batches with no more than five configurations in each batch.

The shoelace formula in Eq. (25) can be generalized to multi-component systems by vectorizing the compositions and energies of arbitrary configurations. For an alloy system with n components, the convex hulls formed by the configurations of such systems are higher-dimension polytopes that cannot be easily visualized. The hyper-volume of the N -D convex hull polytopes can be calculated by summing the volumes of Delauney-triangularized N -D simplices:

$$V = \sum_s V_s \quad (30)$$

in which V_s is the volume of an n -dimensional simplex, such as the triangle in 2D, and the tetrahedron in 3D. The $(n + 1)$ vertices of the simplex are on-hull configurations such that each configuration can be described by an n -dimensional vector, $v_i = [x_i^1, x_i^2, x_i^3, \dots, x_i^{n-1}, E_i]$, in which x_i^j is the concentration of the j -th component and E_i is the energy. Using this vector, the volume of the N -D simplex can be expressed by:

$$V_s = \frac{1}{n!} |\det[A_s]| = \frac{1}{n!} |\det[(v_2 - v_1) \ (v_3 - v_1) \ \dots \ (v_n - v_1) \ (v_{n+1} - v_1)]| \quad (31)$$

in which the $(n \times n)$ matrix A_s is constructed by the coordinates of configurations which occupy the vertices of the simplex V_s . Each column in A_s represents a vector pointing from the reference vertex (noted as v_1) to another vertex (noted as $v_2, v_3, \dots, v_n$). It is noted that only the last row of A_s contains the energy terms E_i ; therefore, the result from the determinant of A_s is lineally related to E_i . This linear relationship can be employed to propagate the uncertainties of energies to the convex hull volumes by plugging Eq. (31) into Eq. (30). Because an on-hull configuration is shared by multiple Delauney-triangularized simplices, simple addition will lead to double-counting the uncertainty of a configuration. To avoid this problem, we need to combine the like terms containing the same E_i by opening the absolute bracket in Eq. (31). The summation will lead to a linear relationship similar to Eq. (25):

$$V = \sum_i f_i(\{x\}) E_i = F * E \quad (32)$$

where $f_i(\{x\})$ represents a set of coefficient functions corresponding to E_i , which can also be expressed in the vector form. These coefficient functions only depend on the compositions of the configurations. For a two-dimensional convex hull, Eq. (33) is reduced to Eq. (25) that $f_i(\{x\}) = \Delta x_i$. And finally, the coefficient function vector can be used to compute the variance of the total volume:

$$\nu_V = F * K(E_i, E_j) * F^T \quad (33)$$

where $K(E_i, E_j)$ is the covariance matrix of energies. With this formulation, the expected improvement policy based on Eq. (23) - Eq. (29) can be generalized to hull volume for n -component systems, leading to the following equation:

$$EI(\tilde{V}, \{\tilde{\sigma}, x\}) = (V_N - \mu_V(\tilde{\sigma}(x))) CDF\left(\frac{V_N - \mu_V(\tilde{\sigma}(x))}{\nu_V(\tilde{\sigma}(x))}\right) + \nu_V(\tilde{\sigma}(x)) PDF\left(\frac{V_N - \mu_V(\tilde{\sigma}(x))}{\nu_V(\tilde{\sigma}(x))}\right) \quad (34)$$

This new metric based on the expected improvement policy is not restricted to the Euclidean space. For example, the compositions of a ternary system can be presented in both orthonormal and the barycentric coordinate

systems; therefore, further studies are required to assess the convex hull volumes calculated in two coordinate systems.

Cluster expansion convex hull based on the genetic algorithms

Genetic algorithms have been used to learn the ground-state energy of binary alloys^{5,7,21}. In a typical workflow employing the genetic algorithms, cluster expansion models are trained by the genetic algorithms²¹. In order to iteratively acquire new configurations for DFT calculations, cross-validation is required to evaluate the best model^{5,7}. This process seeks to find the best set of basis functions to fit cluster expansion with linear regression²¹. First, a candidate model can be described by a string of zeros and ones. A value of one will include the terms of the basis function, and the corresponding J_i will be fitted, while a value of zero will exclude the term in the fitting. A population of candidate models will be ranked according to the cross-validation (CV) scores, which evaluate the predictive power of the candidate models through computing the root mean square errors of the fitting^{35,36}. The evolutionary process picks two candidate models as parent models to produce a child model through the crossover operation and the mutation operation²¹. In the crossover operation, the child model inherits the 'genes' (the string of zeros and ones) from the parent models. Then the child model can mutate by flipping the one and zero values to include/exclude the basis function terms. The child model will be accepted in the population if its CV score improves the current population. The iterative evolution process ends when the child models are not likely to improve the population after some generations. Hart et al. have shown that genetic algorithms may converge within 150 generations for different alloying systems²¹. In our application, 500 generations are used for each iteration. With the best candidate in the population, the convex hull will be predicted, and the on-hull configurations will be selected to be calculated with DFT. The iterative data acquisition process stops when the computation budget is reached, or the ground-state line is found. For convenience, this method is called GA-CE-hull.

Data availability

The Co-Ni system data used in this study is available in the study⁷, and the Zr-O system data in this study is available in the study⁵. The Ni-Al-Cr ternary data is shared by Dr. John Goiri and Prof. Anton Van der Ven. The BG models and acquisition tools are available at https://github.com/Dongsheng-Wen/BGO-CE_Acquisition_Tools.

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Author contributions

D.W.: Conceptualization, Methodology, Calculation, Validation, Visualization, Data Curation, Writing - Original Draft, Writing - Review Editing. V.T.: Validation, Data Curation, Calculation. M.T.: Conceptualization, Methodology, Resources, Writing - Review Editing, Supervision, Project administration.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Dongsheng Wen.

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